

Success Metrics & KPI Framework

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Project: MsingiPACK Academy CRO & Analytics Framework

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1. Purpose

This document defines the KPI framework for measuring the performance of the MsingiPACK Academy acquisition funnel, registration process, user activation, purchasing behavior, and marketing attribution.

The framework separates **business outcomes**, **funnel performance**, **product activation**, and **measurement integrity** so that improvements can be evaluated using reliable and clearly defined metrics.

A key principle is that a KPI should have:

- A clearly defined numerator and denominator
- A consistent data source
- A documented baseline
- A realistic target
- A measurement method that can be reproduced

2. Primary Business KPIs

These metrics measure whether improvements to the acquisition and conversion funnel ultimately produce better commercial outcomes.

KPI	Current Baseline	Target	Definition
Session-to-Purchase Rate	0.45%	1.50%	Valid purchases ÷ GA4 sessions
Average Order Value (AOV)	KES 773.32	KES 1,000	Revenue ÷ valid purchases
Customer Acquisition Cost (CAC)	KES 489.52	< KES 350	Ad spend ÷ valid purchases
Return on Ad Spend (ROAS)	1.58x	3.50x	Verified revenue ÷ ad spend

Baseline calculation

Session-to-Purchase Rate

25 valid payments ÷ 5,611 GA4 sessions = **0.45%**

AOV

KES 19,333 revenue ÷ 25 payments = **KES 773.32**

CAC

KES 12,238 ad spend ÷ 25 payments = **KES 489.52**

ROAS

KES 19,333 revenue ÷ KES 12,238 ad spend = **1.58x**

Measurement consideration

The 0.45% figure should be described as a **Session-to-Purchase Rate**, rather than a generic "funnel conversion rate." Sessions are not equivalent to unique customers, so the denominator should remain explicit.

Where reliable user-level data becomes available, a separate **User-to-Purchase Conversion Rate** should also be reported.

3. Micro-Conversions & Activation Metrics

Business outcomes alone do not explain where users are being lost. These metrics identify the major points of friction between acquisition and purchase.

KPI	Current Baseline	Target	Definition
Ad Intent Clicks	7,486	Improve quality	Meta-reported tracked intent/CTA clicks
Ad Click-to-Account Rate	46.11%	70.00%	Verified registrations ÷ ad CTA clicks
Post-Signup Activation Rate	7.79%	40.00%	Activated users ÷ registrations
Activated-User-to-Purchase Rate	9.29%	15.00%	Purchases ÷ activated users

3.1 Ad Intent Clicks

Meta recorded **7,486 intent/CTA clicks**.

The previous framework described this as a 100% "Ad Click-to-Form Initiation Rate." This is not technically a conversion rate because the same event is being used as both the tracked action and the denominator.

Therefore, the metric should instead be treated as:

Tracked Ad Intent Clicks = 7,486

A true click-to-form initiation rate should only be introduced when an independent **form initiation event** is available.

3.2 Ad Click-to-Account Rate

Baseline: 46.11%

Calculation:

3,452 verified database accounts ÷ 7,486 ad CTA clicks = **46.11%**

Target

70.00%

The objective is to reduce registration friction and ensure that users arriving from paid campaigns can successfully complete account creation.

Potential intervention areas include:

- Registration form usability
 - Mobile UX
 - Form validation
 - Required-field friction
 - Authentication errors
 - Cross-domain/session continuity
 - Event implementation
 - Failed registration tracking
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4. Post-Signup Activation

Current baseline: 7.79%

Calculation:

269 active/logged-in users ÷ 3,452 registered accounts = **7.79%**

Target

40.00%

The proposed target assumes that onboarding improvements can substantially increase the number of registered users who actually return to and engage with the Academy.

Potential interventions include:

- Automated SMS onboarding
- WhatsApp reminders
- Welcome messaging
- Clear next-step instructions
- Direct links to the learner dashboard
- Course-start prompts
- Recovery campaigns for inactive registrants

Important measurement refinement

"At least one login" is a relatively weak definition of activation.

For future reporting, activation should preferably represent a meaningful product action, for example:

Activated User = Registered user who logs in and reaches the first meaningful learning activity or course page.

This prevents a login event from being treated as proof that a user genuinely started using the product.

5. Activated-User-to-Purchase Rate

Current baseline: 9.29%

Calculation:

25 valid payments ÷ 269 activated users = **9.29%**

Target

15.00%

This metric measures the ability of the Academy to convert users who have already demonstrated meaningful engagement into paying customers.

It should be treated separately from the overall session-to-purchase rate because it represents a much later stage of the funnel.

6. Measurement Integrity KPIs

A Growth Engineering framework should not only measure conversion. It must also establish whether the underlying measurement system can be trusted.

6.1 Attribution Classification Gap

Current observation: 74.50% of traffic identified as mislabeled/problematic

Target

<5.00% unassigned or incorrectly classified traffic

This metric should distinguish between:

1. **Unassigned traffic**
2. **Incorrectly classified traffic**
3. **Paid traffic appearing as Paid Other**
4. **Campaign traffic losing UTM parameters**
5. **Traffic incorrectly attributed to Direct**

The objective is not simply to reduce the percentage labelled "Paid Other." The objective is to ensure that campaign traffic is consistently classified according to its true acquisition source.

Measurement method

Compare expected campaign source/medium information against GA4 channel classification.

Primary source:

- GA4
 - GTM
 - Campaign parameters
 - Cross-domain attribution data
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7. Purchase Event Integrity

7.1 Purchase Event Multiplication

Observed baseline: 8.25 purchase events per buyer

Calculation:

33 purchase events ÷ 4 unique buyers = **8.25 events per buyer**

This should **not** be labelled an "8.25x duplicate rate."

The figure represents the average number of purchase events being generated for each buyer.

Target

1.00 purchase event per transaction

The objective is to ensure that each genuine transaction generates exactly one purchase event.

Measurement method

Compare:

Total purchase events

against

Unique transaction IDs

A healthy implementation should approach:

1 unique purchase event = 1 valid transaction

Duplicate events can distort:

- GA4 revenue
- Conversion reporting
- Meta optimization
- ROAS
- Campaign performance
- Automated bidding
- Executive reporting

8. KPI Measurement Framework

Metric	Current Baseline	Target	Measurement Method	Primary Data Source
Session-to-Purchase Rate	0.45%	1.50%	Valid purchases ÷ GA4 sessions	GA4 + M-PESA Ledger
Ad Intent Clicks	7,486	Improve quality	Tracked Meta intent clicks	Meta Ads
Ad Click-to-Account Rate	46.11%	70.00%	Verified registrations ÷ ad CTA clicks	Moodle DB + GTM
Post-Signup Activation	7.79%	40.00%	Activated users ÷ registrations	Moodle DB + User Logs

Metric	Current Baseline	Target	Measurement Method	Primary Data Source
Activated-User-to-Purchase Rate	9.29%	15.00%	Valid purchases ÷ activated users	Moodle + M-PESA
Attribution Classification Gap	74.50% problematic	<5.00%	Incorrect/unassigned sessions ÷ total sessions	GA4
Purchase Event Multiplication	8.25x	1.00x	Purchase events ÷ unique transactions	GA4 + GTM
AOV	KES 773.32	KES 1,000	Revenue ÷ valid purchases	M-PESA Ledger
CAC	KES 489.52	< KES 350	Ad spend ÷ valid purchases	Meta Ads + Ledger
ROAS	1.58x	3.50x	Verified revenue ÷ ad spend	M-PESA Ledger + Meta Ads

9. Target Consistency Check

The CAC and ROAS targets should be interpreted together rather than independently.

At a target AOV of **KES 1,000**:

If CAC = KES 350

$\text{KES 1,000} \div \text{KES 350} = \text{approximately } \mathbf{2.86x \text{ ROAS}}$

Therefore, a KES 350 CAC target does **not** independently imply a 3.50x ROAS.

To achieve 3.50x ROAS at KES 1,000 AOV

$\text{KES 1,000} \div 3.50 = \text{approximately } \mathbf{\text{KES 286 CAC}}$

Therefore, the KPI framework should recognize:

KES 350 CAC and 3.50x ROAS are not mathematically equivalent targets.

If 3.50x ROAS is the strategic target, the implied CAC at KES 1,000 AOV should be approximately **KES 286 or lower**.

10. AOV Target Consideration

The current AOV is:

KES 773.32

The proposed target is:

KES 1,000

This represents approximately a **29.3% increase**.

However, if the core Academy product has a substantially lower individual purchase price, achieving KES 1,000 AOV may require additional commercial mechanisms such as:

- Bundles
- Multiple-course purchases
- Upsells
- Premium products
- Subscription plans
- Additional learning products

Therefore, AOV should not be treated purely as a CRO metric unless the product architecture provides a realistic path to increasing transaction value.

11. KPI Hierarchy

The overall measurement framework can be organized into four layers.

Layer 1 — Business Outcomes

Revenue → Purchases → AOV → CAC → ROAS

These determine whether the business is generating economically valuable customers.

Layer 2 — Funnel Conversion

Ad Click → Registration → Activation → Purchase

These identify where potential customers are being lost.

Layer 3 — Product Activation

Registration → Login → Learning Activity → Purchase

These determine whether acquired users actually begin using the Academy.

Layer 4 — Measurement Integrity

Attribution → Event Accuracy → Transaction Deduplication → Cross-Domain Continuity

These determine whether the numbers being used to make decisions can be trusted.

12. Recommended Reporting Cadence

Weekly

Monitor:

- Sessions
- Ad clicks
- Registrations
- Activation
- Purchases
- Conversion rates
- Purchase-event duplication
- Attribution anomalies

Monthly

Evaluate:

- Revenue
- AOV
- CAC
- ROAS
- Channel performance
- Funnel leakage
- Activation performance
- Measurement quality

After Major Tracking Changes

Perform a measurement validation before comparing new results with historical data.

This is particularly important when changing:

- GTM triggers
 - GA4 events
 - Meta Pixel events
 - Cross-domain tracking
 - Purchase tracking
 - Transaction IDs
 - Attribution parameters
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13. Success Definition

The MsingiPACK Academy optimization initiative should not be considered successful simply because individual GA4 or Meta metrics increase.

Success requires improvement across three dimensions:

1. Commercial Performance

- Higher purchase conversion
- Higher revenue
- Lower CAC
- Higher ROAS
- Sustainable AOV improvement

2. User Journey Performance

- Higher registration completion
- Higher activation
- Lower funnel abandonment
- Higher activated-user purchase conversion

3. Measurement Reliability

- Correct campaign attribution
- Reliable cross-domain tracking
- One purchase event per transaction
- Reconciliation between GA4, Moodle, Meta and the M-PESA ledger

The final objective is therefore:

Build a measurement system that accurately represents the customer journey and use that reliable data to identify and reduce revenue leakage throughout the acquisition-to-purchase funnel.

This makes the KPI framework both a **business performance framework** and a **Growth Engineering measurement specification**.